

# Week 2: Study Sheet

## EE0849 Introduction to Robotics - Problems & Solutions

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# 1 Composition of Rotations

## 1.1 Problem 1.1

Find the final rotation matrix for the following sequence:

1. Rotate  $90^\circ$  about the **current** z-axis
2. Rotate  $90^\circ$  about the **current** x-axis

### 1.1.1 Solution

For rotations about the **current (body) frame**, we post-multiply (multiply on the right):

$$R_{final} = R_z(90^\circ) \cdot R_x(90^\circ)$$

$$R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_x(90^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$R_{final} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$


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## 1.2 Problem 1.2

Find the final rotation matrix for:

1. Rotate  $90^\circ$  about the **fixed** z-axis
2. Rotate  $90^\circ$  about the **fixed** x-axis

### 1.2.1 Solution

For rotations about the **fixed (world) frame**, we pre-multiply (multiply on the left):

$$R_{final} = R_x(90^\circ) \cdot R_z(90^\circ)$$

Note the reversed order compared to Problem 1.1!

$$R_{final} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

**Key insight:** Different result than Problem 1.1! Order matters.

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### 1.3 Problem 1.3

A robot arm performs:

1. Rotate  $45^\circ$  about **fixed** z-axis
2. Rotate  $30^\circ$  about **current** y-axis

Find  $R_{final}$ .

#### 1.3.1 Solution

- Step 1 (fixed z): Pre-multiply
- Step 2 (current y): Post-multiply

Starting with  $R = I$ :

- After step 1:  $R = R_z(45^\circ) \cdot I = R_z(45^\circ)$
- After step 2:  $R = R_z(45^\circ) \cdot R_y(30^\circ)$

With  $c_{45} = s_{45} = \frac{\sqrt{2}}{2} \approx 0.707$ ,  $c_{30} = \frac{\sqrt{3}}{2} \approx 0.866$ ,  $s_{30} = 0.5$ :

$$R_z(45^\circ) = \begin{bmatrix} 0.707 & -0.707 & 0 \\ 0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_y(30^\circ) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0 & 1 & 0 \\ -0.5 & 0 & 0.866 \end{bmatrix}$$

$$R_{final} = \begin{bmatrix} 0.612 & -0.707 & 0.354 \\ 0.612 & 0.707 & 0.354 \\ -0.5 & 0 & 0.866 \end{bmatrix}$$

---

## 2 Homogeneous Transformations

### 2.1 Problem 2.1

Frame  $\{1\}$  is rotated  $90^\circ$  about z-axis and translated by  $d = [2, 1, 0]^T$  relative to frame  $\{0\}$ . Find  $H_1^0$ .

#### 2.1.1 Solution

$$H_1^0 = \begin{bmatrix} R_1^0 & d_1^0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_z(90^\circ) & d \\ 0 & 1 \end{bmatrix}$$

$$H_1^0 = \begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

---

### 2.2 Problem 2.2

Using  $H_1^0$  from Problem 2.1, transform point  $p^1 = [1, 0, 0]^T$  to frame  $\{0\}$ .

### 2.2.1 Solution

$$\begin{aligned} \begin{bmatrix} p^0 \\ 1 \end{bmatrix} &= H_1^0 \begin{bmatrix} p^1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \cdot 1 + (-1) \cdot 0 + 0 \cdot 0 + 2 \\ 1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 1 \\ 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 + 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

**Answer:**  $p^0 = [2, 2, 0]^T$

---

### 2.3 Problem 2.3

Given:

$$H_1^0 = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H_2^1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Find  $H_2^0$  and the position of origin of  $\{2\}$  in  $\{0\}$ .

#### 2.3.1 Solution

$$\begin{aligned} H_2^0 &= H_1^0 \cdot H_2^1 = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -1 & 0 & 3 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

**Origin of  $\{2\}$  in  $\{0\}$ :** Read from last column =  $[3, 2, 0]^T$

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## 3 Inverse of Homogeneous Transforms

### 3.1 Problem 3.1

Find the inverse of:

$$H = \begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

### 3.1.1 Solution

**Formula:**  $H^{-1} = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix}$

Extract  $R$  and  $d$ :

$$R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad d = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Compute  $R^T$ :

$$R^T = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Compute  $-R^T d$ :

$$-R^T d = - \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = - \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}$$

$$H^{-1} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

**Verify:**  $H \cdot H^{-1} = I_4$

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## 3.2 Problem 3.2

For pure translation  $H = \begin{bmatrix} I & d \\ 0 & 1 \end{bmatrix}$ , what is  $H^{-1}$ ?

### 3.2.1 Solution

Since  $R = I$ , we have  $R^T = I$ .

$$-R^T d = -I \cdot d = -d$$

$$H^{-1} = \begin{bmatrix} I & -d \\ 0 & 1 \end{bmatrix}$$

**For pure translation, just negate the translation vector!**

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## 4 Euler Angles

### 4.1 Problem 4.1

Given Z-Y-Z Euler angles  $\phi = 90^\circ$ ,  $\theta = 90^\circ$ ,  $\psi = 0^\circ$ , find the rotation matrix.

#### 4.1.1 Solution

$$R_{ZYZ} = R_z(\phi) \cdot R_y(\theta) \cdot R_z(\psi)$$

$$R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_y(90^\circ) = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \quad R_z(0^\circ) = I$$

$$R_{ZYZ} = R_z(90^\circ) \cdot R_y(90^\circ) \cdot I = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}$$


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#### 4.2 Problem 4.2

Extract Z-Y-Z Euler angles from:

$$R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

#### 4.2.1 Solution

For Z-Y-Z Euler angles, we use:

- $r_{13} = c_\phi s_\theta = 1$
- $r_{23} = s_\phi s_\theta = 0$
- $r_{33} = c_\theta = 0$
- $r_{31} = -s_\theta c_\psi = -1$
- $r_{32} = s_\theta s_\psi = 0$

From  $r_{33} = 0$ :  $\theta = 90^\circ$

Since  $\sin \theta = 1 \neq 0$  (no gimbal lock):

$$\phi = \text{atan2}(r_{23}, r_{13}) = \text{atan2}(0, 1) = 0^\circ$$

$$\psi = \text{atan2}(r_{32}, -r_{31}) = \text{atan2}(0, 1) = 0^\circ$$

**Answer:**  $\phi = 0^\circ, \theta = 90^\circ, \psi = 0^\circ$

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#### 4.3 Problem 4.3 (Gimbal Lock)

Extract Z-Y-Z Euler angles from:

$$R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

### 4.3.1 Solution

From  $r_{33} = 1$ :  $\cos \theta = 1 \Rightarrow \theta = 0^\circ$

**Gimbal lock condition!** When  $\theta = 0$ , the  $z$  and  $z'$  axes align.

We can only determine  $\phi + \psi$ , not individual values.

From the matrix (which is  $R_z(90^\circ)$ ):

$$\phi + \psi = 90^\circ$$

Any combination works:  $(\phi = 90^\circ, \psi = 0^\circ)$ ,  $(\phi = 45^\circ, \psi = 45^\circ)$ , etc.

**Convention:** Often set  $\psi = 0$ , so  $\phi = 90^\circ$ .

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## 5 Roll-Pitch-Yaw

### 5.1 Problem 5.1

An aircraft has Roll =  $30^\circ$ , Pitch =  $20^\circ$ , Yaw =  $45^\circ$ . Find the rotation matrix.

#### 5.1.1 Solution

Roll-Pitch-Yaw uses fixed axes X-Y-Z, so we pre-multiply in reverse order:

$$R_{RPY} = R_z(\psi) \cdot R_y(\theta) \cdot R_x(\phi) = R_z(45^\circ) \cdot R_y(20^\circ) \cdot R_x(30^\circ)$$

With  $c_{45} = s_{45} = 0.707$ ,  $c_{20} = 0.940$ ,  $s_{20} = 0.342$ ,  $c_{30} = 0.866$ ,  $s_{30} = 0.5$ :

$$R_x(30^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{bmatrix}$$

$$R_y(20^\circ) = \begin{bmatrix} 0.940 & 0 & 0.342 \\ 0 & 1 & 0 \\ -0.342 & 0 & 0.940 \end{bmatrix}$$

$$R_z(45^\circ) = \begin{bmatrix} 0.707 & -0.707 & 0 \\ 0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_{RPY} = \begin{bmatrix} 0.665 & -0.612 & 0.429 \\ 0.665 & 0.747 & -0.006 \\ -0.342 & 0.259 & 0.903 \end{bmatrix}$$

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### 5.2 Problem 5.2

Extract Roll-Pitch-Yaw angles from the rotation matrix computed in Problem 5.1.

### 5.2.1 Solution

**RPY extraction formulas** (given  $R = [r_{ij}]$ ):

$$\theta = \text{atan2}\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right) \quad (\text{pitch})$$

$$\phi = \text{atan2}(r_{32}, r_{33}) \quad (\text{roll})$$

$$\psi = \text{atan2}(r_{21}, r_{11}) \quad (\text{yaw})$$

Applying to  $R_{RPY}$  from Problem 5.1 (using exact trig values, not rounded entries):

- Pitch:  $\theta = \text{atan2}(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}) = \text{atan2}(s_{20}, c_{20}) = 20^\circ$
- Roll:  $\phi = \text{atan2}(r_{32}, r_{33}) = \text{atan2}(c_{20}s_{30}, c_{20}c_{30}) = \text{atan2}(s_{30}, c_{30}) = 30^\circ$
- Yaw:  $\psi = \text{atan2}(r_{21}, r_{11}) = \text{atan2}(s_{45}c_{20}, c_{45}c_{20}) = \text{atan2}(s_{45}, c_{45}) = 45^\circ$

**Answer:** Roll =  $30^\circ$ , Pitch =  $20^\circ$ , Yaw =  $45^\circ$  (recovers Problem 5.1 inputs).

**Gimbal lock** occurs when  $\cos \theta = 0$  ( $\theta = \pm 90^\circ$ ). In that case, only  $\phi \pm \psi$  is determined.

## 6 Unit Quaternions

### 6.1 Problem 6.1

Convert the rotation defined by axis  $\hat{k} = \frac{1}{\sqrt{3}}[1, 1, 1]^T$  and angle  $\theta = 60^\circ$  to a unit quaternion.

#### 6.1.1 Solution

**Axis-angle to quaternion:**

$$\mathbf{q} = \left( \cos \frac{\theta}{2}, \sin \frac{\theta}{2} \hat{k} \right) = (w, x, y, z)$$

With  $\theta = 60^\circ$ :

$$w = \cos 30^\circ = \frac{\sqrt{3}}{2} \approx 0.866$$

$$\sin 30^\circ = 0.5, \quad \sin \frac{\theta}{2} \hat{k} = 0.5 \cdot \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.289 \\ 0.289 \\ 0.289 \end{bmatrix}$$

$$\mathbf{q} = (0.866, 0.289, 0.289, 0.289)$$

**Verify:**  $\|\mathbf{q}\| = \sqrt{0.866^2 + 3 \times 0.289^2} = \sqrt{0.75 + 0.25} = 1$

### 6.2 Problem 6.2

What quaternion represents the identity rotation ( $R = I$ )?

### 6.2.1 Solution

The identity rotation has  $\theta = 0$ , so:

$$w = \cos 0 = 1, \quad (x, y, z) = \sin 0 \cdot \hat{k} = (0, 0, 0)$$

$$\mathbf{q}_{\text{identity}} = (1, 0, 0, 0)$$

Note:  $\mathbf{q}$  and  $-\mathbf{q}$  represent the **same rotation** (double cover of  $SO(3)$ ).

## 7 Summary of Key Formulas

| Concept                    | Formula                                                                        |
|----------------------------|--------------------------------------------------------------------------------|
| Current frame rotation     | Post-multiply: $R_{\text{new}} = R_{\text{current}} \cdot R_{\text{rotation}}$ |
| Fixed frame rotation       | Pre-multiply: $R_{\text{new}} = R_{\text{rotation}} \cdot R_{\text{current}}$  |
| Homogeneous transform      | $H = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix}$                             |
| Transform composition      | $H_2^0 = H_1^0 \cdot H_2^1$                                                    |
| Transform inverse          | $H^{-1} = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix}$                 |
| Z-Y-Z Euler angles         | $R = R_z(\phi) R_y(\theta) R_z(\psi)$                                          |
| Roll-Pitch-Yaw             | $R = R_z(\psi) R_y(\theta) R_x(\phi)$                                          |
| RPY extraction             | $\theta = \text{atan2}(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2})$ , etc.            |
| Quaternion from axis-angle | $\mathbf{q} = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \hat{k})$          |

## 8 Reference Formulas

### 8.1 Z-Y-Z Expanded Matrix

$$R_{ZYZ} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

where (using shorthand  $c_\phi = \cos \phi$ ,  $s_\phi = \sin \phi$ , etc.):

$$r_{11} = c_\phi c_\theta c_\psi - s_\phi s_\psi, \quad r_{12} = -c_\phi c_\theta s_\psi - s_\phi c_\psi, \quad r_{13} = c_\phi s_\theta$$

$$r_{21} = s_\phi c_\theta c_\psi + c_\phi s_\psi, \quad r_{22} = -s_\phi c_\theta s_\psi + c_\phi c_\psi, \quad r_{23} = s_\phi s_\theta$$

$$r_{31} = -s_\theta c_\psi, \quad r_{32} = s_\theta s_\psi, \quad r_{33} = c_\theta$$

**Extraction formulas** (when  $\sin \theta \neq 0$ ):

$$\theta = \text{atan2}\left(\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right)$$

$$\phi = \text{atan2}(r_{23}, r_{13})$$

$$\psi = \text{atan2}(r_{32}, -r_{31})$$

When  $\sin \theta = 0$  (gimbal lock), only  $\phi + \psi$  (if  $\theta = 0$ ) or  $\phi - \psi$  (if  $\theta = \pi$ ) is uniquely determined.

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## 8.2 RPY Extraction Formulas

Given  $R = [r_{ij}]$ , extract Roll-Pitch-Yaw angles (when  $\cos \theta \neq 0$ ):

$$\theta = \text{atan2}\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right) \quad (\text{pitch})$$

$$\phi = \text{atan2}(r_{32}, r_{33}) \quad (\text{roll})$$

$$\psi = \text{atan2}(r_{21}, r_{11}) \quad (\text{yaw})$$

When  $\cos \theta = 0$  ( $\theta = \pm 90^\circ$ ), gimbal lock occurs and only  $\phi \pm \psi$  is determined.

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## 8.3 Unit Quaternions

A unit quaternion encodes rotation as:

$$\mathbf{q} = \left( \cos \frac{\theta}{2}, \sin \frac{\theta}{2} \hat{k} \right) = (w, x, y, z), \quad \|\mathbf{q}\| = 1$$

**Key properties:**

- $\mathbf{q}$  and  $-\mathbf{q}$  represent the same rotation (double cover)
  - Quaternion inverse:  $\mathbf{q}^{-1} = \mathbf{q}^* = (w, -x, -y, -z)$  for unit quaternions
  - Composition:  $\mathbf{q}_{total} = \mathbf{q}_1 \otimes \mathbf{q}_2$  (Hamilton product)
  - **SLERP** (Spherical Linear Interpolation):  $\text{slerp}(\mathbf{q}_0, \mathbf{q}_1, t) = \mathbf{q}_0 (\mathbf{q}_0^{-1} \mathbf{q}_1)^t$  — smooth, constant-speed interpolation between orientations
  - No singularities (unlike all 3-parameter representations)
- 

## 8.4 Special-Case Homogeneous Transforms

**Pure translation** ( $R = I$ ):

$$H_{trans}(d) = \begin{bmatrix} I & d \\ 0 & 1 \end{bmatrix}, \quad H_{trans}^{-1} = \begin{bmatrix} I & -d \\ 0 & 1 \end{bmatrix}$$

**Pure rotation** ( $d = 0$ ):

$$H_{rot}(R) = \begin{bmatrix} R & 0 \\ 0 & 1 \end{bmatrix}, \quad H_{rot}^{-1} = \begin{bmatrix} R^T & 0 \\ 0 & 1 \end{bmatrix}$$


---

## 8.5 Active vs Passive Interpretation

- **Active (operator) view:** rotate a physical vector in the same frame

$$v' = Rv$$

- **Passive (alias) view:** re-express coordinates between frames

$$p^0 = R_1^0 p^1 \quad \Leftrightarrow \quad p^1 = (R_1^0)^T p^0$$

Same matrix entries, different interpretation. In this course we primarily use the passive view (coordinate transformation between frames).

---

## 8.6 Common Mistakes

1. **Wrong multiplication order** - Fixed frame = pre-multiply, Current frame = post-multiply
2. **Using  $H^T$  as inverse** - Must use the formula with  $-R^T d$
3. **Gimbal lock** - When  $\theta = 0$  or  $180^\circ$  in Euler angles, only sum/difference of angles is determined
4. **Forgetting the “1”** - Homogeneous coordinates require appending 1 to position vectors